Sunwoo **Lee**

Associate Professor
Department of Computer Engineering
Inha University, South Korea

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https://lmls-lab.github.io

Research Interest

- Large-Scale Distributed Machine Learning and Deep Learning
- Federated Learning on Heterogeneous Systems
- Applied Machine Learning (E.g., Physics + ML)

Education

Northwestern University — Evanston, IL USA
Ph.D. in Computer Engineering — 2020
Advisors: Prof. Alok Choudhary and Prof. Wei-keng Liao
Thesis: Scalable Parallelization Strategy for Large-Scale Deep Learning

Hanyang University — Seoul, South Korea
B.S. and M.S. in Computer Engineering — 2009
Advisor: Prof. Minsoo Ryu

Employment

Inha University — Incheon, South Korea
Assistant Professor of Computer Engineering — 2022 – now

University of Southern California — Los Angeles, CA USA
Postdoctoral Researcher — 2020 – 2022
Advisor: Prof. Salman Avestimehr

Samsung Electronics, Memory Solutions Lab. — Hwaseong, South Korea
System Software Researcher — 2013 –2015

Humax — Bundang, South Korea
Software Engineer (Alternative Military Service) — 2009 –2013

Research Internship

Lawrence Berkeley National Laboratory — Berkeley, CA USA
Research Intern — Jun 2020 – Aug 2020

Fermi National Accelerator Laboratory — Batavia, IL USA
Research Intern — Jul 2019 – Sep 2019

Argonne National Laboratory — Lemont, IL USA
Research Intern — Jun 2018 – Aug 2018

Teaching Experience

Inha University
- CSE4317: Deep Learning — Fall 2026 ~ 2026
- CSE4315: Machine Learning — Spring 2023 ~ 2026
- CSE1103: Objected Oriented Programming 2 — Spring 2023 ~ 2024
- CSE1112: Introduction to Computer Engineering — Spring 2024, 2025
- CSE3209: System Programming — Fall 2022, 2023, 2024
- CSE3313: Linux Programming — Fall 2022, 2023, 2024

University of Southern California
- AEOP Scholarship Program in Data Science — Summer 2021

Northwestern University
- CE501: Social Media Mining — Spring 2020
- CE303: Advanced Digital Design — Fall 2019

Honors & Awards

NeurIPS Top Reviewer — 2026
- Recognized as a Top Reviewer and get awarded a complimentary full conference registration.

KDD Outstanding Reviewer — 2025
- Top 10% reviewer among technical program committee

FL-AAAI Workshop Best Paper Award — 2022
- SSFL: Tackling Label Deficiency in Federated Learning via Personalized Self-Supervision

IEEE HiPC Best Paper Finalist — 2017
- Parallel Deep Convolutional Neural Network Training by Exploiting the Overlapping of Computation and Communication

Publications

1. Vincent-Daniel Yun, Junhyuk Jo, Sai Praneeth Karimireddy, **Sunwoo Lee**, Ghosted Layers: Unconstrained Activation Alignment for Recovering Layer-Pruned LLMs, *NeurIPS* 2026

2. Vincent-Daniel Yun, Youngrae Kim, Woosang Lim, YoungJin Heo, Minkyu Kim, **Sunwoo Lee**, Locality-aware Redundancy Pruning for LLM Depth Compression, *EMNLP* 2026

3. Sanghoon Yeom†, Junhyuk Jo†, Haeyun Song, Sung Myung, **Sunwoo Lee***, and Hyungwoo Lee*, Morphology-Embedded Signatures of Lattice Strain in Ferroelectric BaTiO3 Thin Films Revealed by Machine Learning, *Small* 2026

4. Youngeun Kim, Youjia Zhang, Huiling Liu, Aecheon Jung, **Sunwoo Lee**, Sungeun Hong, ZOO-Prune: Training-Free Token Pruning via Zeroth-Order Gradient Estimation in Vision-Language Models, *CVPR* 2026

5. Jihyun Lim, Tuo Zhang, and **Sunwoo Lee**, Multi-Metric Client Activation Method for Fast and Accurate Federated Learning, *ACM Transactions on Intelligent Systems and Technology (TIST)*, **47**, 1-27 2026

6. Sungmin Kang, Jisoo Kim, and **Sunwoo Lee**, GEM: A Scale-Aware and Distribution-Sensitive Sparse Fine-Tuning Framework for Effective Downstream Adaptation, *AAAI Conference on Artificial Intelligence (AAAI)* 2026

7. Jisoo Kim, Sungmin Kang, and **Sunwoo Lee**, Layer-wise Update Aggregation with Recycling for Communication-Efficient Federated Learning, *Annual Conference on Neural Information Processing Systems (NeurIPS)* 2025

8. Jihyun Lim, Junhyuck Jo, Tuo Zhang, and **Sunwoo Lee**, Enabling Weak Client Participation via On-Device Knowledge Distillation in Heterogeneous Federated Learning, *European Conference on Artificial Intelligence (ECAI)* 2025

9. Tuo Zhang, Tiantian Feng, Dimitrios Dimitriadis, **Sunwoo Lee**, Mi Zhang, Shrikanth S. Narayanan, Salman Avestimehr, GPT-FL: Generative Pre-Trained Model-Assisted Federated Learning, *FedVision workshop held conjunction with CVPR* 2025

10. Doyeop Kim*, Jung-Woo Lee*, Jihyun Lim, Sungjun Choi, Khimananda Acharya, Seobin Oh, Jaewhan Oh, Tula R. Paudel, Yongsoo Yang, Kitae Eom*, **Sunwoo Lee***, and Hyungwoo Lee*, Highly Stable Two-level Current Fluctuation in Complex Oxide Heterostructures, *Nature Communications*, 2025

11. Sanghyeok Ryou, Jihyun Lim, Minwoo Jang, Kitae Eom, **Sunwoo Lee***, and Hyungwoo Lee*, Machine Learning Approach to Characterize Ferromagnetic La0.7Sr0.3MnO3 Thin Films via Featurization of Surface Morphology, *Advanced Science*, 2025

12. **Sunwoo Lee**, Layer-Wise Adaptive Gradient Norm Penalizing Method for Efficient and Accurate Deep Learning, *ACM SIGKDD* 2024

13. **Sunwoo Lee**, Tuo Zhang, Saurav Prakash, Yue Niu, and Salman Avestimehr, Embracing Federated Learning: Enabling Weak Client Participation via Partial Model Training, *IEEE Transactions on Mobile Computing* 2024

14. **Sunwoo Lee**, Anit Sahu, Chaoyang He, and Salman Avestimehr, Partial Model Averaging in Federated Learning: Performance Guarantees and Benefits, *Neurocomputing* 2023

15. Yue Niu, Saurav Prakash, Souvik Kundu, **Sunwoo Lee**, and Salman Avestimehr, Overcoming Resource Constraints in Federated Learning: Large Models Can Be Trained with only Weak Clients, *Transactions on Machine Learning Research* 2023

16. Yue Niu, Zalan Fabian, **Sunwoo Lee**, Mahdi Soltanolkotabi, and Salman Avestimehr, mL-BFGS: A Momentum-based L-BFGS for Distributed Large-Scale Neural Network Optimization, *Transactions on Machine Learning Research* 2023

17. **Sunwoo Lee,** Tuo Zhang, and Salman Avestimehr, Layer-wise Adaptive Model Aggregation for Scalable Federated Learning, *AAAI Conference on Artificial Intelligence (AAAI),* **oral presentation** (19.7%) 2023

18. Tuo Zhang, TianTian Feng, Samiul Alam, **Sunwoo Lee**, Mi Zhang, Shrikanth S. Narayanan, and Salman Avestimehr, FedAudio: A Federated Learning Benchmark for Audio Tasks, *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)* 2023

19. **Sunwoo Lee**, Chaoyang He, and Salman Avestimehr, Achieving Small-Batch Accuracy with Large-Batch Scalability via Hessian-Aware Learning Rate Adjustment. *Elsevier Neural Networks*, **158**, 1-14 2023

20. **Sunwoo Lee**, Jaeyong Jeon, and Hyungwoo Lee, Probing Oxygen Vacancy Distribution in Oxide Heterostructures by Deep Learning-based Spectral Analysis of Current Noise. *Applied Surface Science*, p154599 2022

21. **Sunwoo Lee**, Jaeyoung Jeon, Kitae Eom, Chaehwa Jeong, Yongsoo Yang, Ji-Yong Park, Chang Beom Eom, and Hyungwoo Lee, Variance-aware Weight Quantization of Multi-level Resistive Switching Devices based on Pt/LaAlO3/SrTiO3 Heterostructures. *Scientific Reports*, **12**, 1-10 2022

22. Kewei Wang, **Sunwoo Lee**, Jan Balewski, Alex Sim, Peter Nugent, Ankit Agrawal, Alok Choudhary, Kesheng Wu, and Wei-keng Liao, Using Multi-resolution Data to Accelerate Neural Network Training in Scientific Applications. *International Symposium on Cluster, Cloud and Internet Computing (CCGrid)* 2022

23. **Sunwoo Lee**, Qiao Kang, Reda Al-Bahrani, Ankit Agrawal, Alok Choudhary, and Wei-keng Liao, Improving Scalability of Parallel CNN Training by Adaptively Adjusting 2022

Parameter Update Frequency. *Journal of Distributed and Parallel Computing*, **159**, 10-23

24. **Sunwoo Lee**, Kai-yuan Hou, Kewei Wang, Saba Sehrish, Marc Paterno, James Kowalkowski, Quincey Koziol, Ross Robert, Ankit Agrawal, Alok Choudhary, and Wei-keng Liao, A Case Study on Parallel HDF5 Dataset Concatenation for High-Energy Physics Data Analysis. *Parallel Computing*, **110**, 102877 — 2022

25. Kai-yuan Hou, Qiao Kang, **Sunwoo Lee**, Ankit Agrawal, Alok Choudhary, and Wei-keng Liao, Supporting Data Compression in PnetCDF, *International Conference on BigData* (19.9%) — 2021

26. **Sunwoo Lee**, Qiao Kang, Kewei Wang, Jan Balewski, Alex Sim, Kesheng Wu, Ankit Agrawal, Alok Choudhary, Peter Nugent, and Wei-keng Liao, Asynchronous I/O Strategy for Large-Scale Deep Learning Applications. *International Conference on High-Performance Computing, Data, and Analytics (HiPC)* (22.9%) — 2021

27. Reda Al-bahrani, Dipendra Jha, Qiao Kang, **Sunwoo Lee**, Zijiang Yang, Wei-keng Liao, Ankit Agrawal, and Alok Choudhary, SIGRNN: Synthetic minority Instances Generation in imbalanced datasets using a Recurrent Neural Network. *International Conference on Pattern Recognition Applications and Methods* — 2021

28. **Sunwoo Lee**, Qiao Kang, Ankit Agrawal, Alok Choudhary, and Wei-keng Liao, Communication-Efficient Local SGD for Scalable Deep Learning. *International Conference on Big Data* (15.7%) — 2020

29. Sandeep Madireddy, Ji Hwan Park, **Sunwoo Lee**, Prasanna Balaprakash, Shinjae Yoo, Wei-keng Liao, Cory Hauck, M. Paul Laiu, and Richard Archibald, In Situ Compression Artifact Removal in Scientific Data Using Deep Transfer Learning and Experience Replay. *Machine Learning: Science and Technology*, **2**, 025010 — 2020

30. Qiao Kang, **Sunwoo Lee**, Ankit Agrawal, Alok Choudhary, and Wei-keng Liao, Improving All-to-many Personalized Communication in MPI I/O. *International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)* — 2020

31. Qiao Kang, **Sunwoo Lee**, Kai-yuan Hou, Robert Ross, Ankit Agrawal, Alok Choudhary, and Wei-keng Liao, Improving MPI Collective I/O for High Volume Non-contiguous Requests with Intra-node Aggregation. *IEEE Transactions on Parallel and Distributed Systems*, 31, 11, 2682-2695 — 2020

32. Qiao Kang, Alex Sim, Peter Nugent, **Sunwoo Lee**, Wei-keng Liao, Ankit Agrawal, Alok Choudhary, and Kesheng Wu. Predicting Resource Requirement in Intermediate Palomar Transient Factory Workflow. *International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, 2020 — 2020

33. **Sunwoo Lee**, Qiao Kang, Sandeep Madireddy, Prasanna Balaprakash, Ankit Agrawal, Alok Choudhary, Richard Archibald, and Wei-keng Liao. Improving Scalability of Parallel CNN Training by Adjusting Mini-Batch Size at Run-Time. *International Conference on Big Data* (18.7%) — 2019

34. **Sunwoo Lee**, Ankit Agrawal, Prasanna Balaprakash, Alok Choudhary, and Wei-keng Liao. Communication-Efficient Parallelization Strategy for Deep Convolutional Neural Network Training. *Machine Learning in High-Performance Computing Environments (MLHPC)* 2018

35. **Sunwoo Lee**, Dipendra Jha, Ankit Agrawal, Alok Choudhary, and Wei-keng Liao. Parallel Deep Convolutional Neural Network Training by Exploiting the Overlapping of Computation and Communication. *International Conference on High-Performance Computing, Data, and Analytics (HiPC)* (22.8%) 2017

36. **Sunwoo Lee**, Wei-keng Liao, Ankit Agrawal, Nikos Hardavellas, and Alok Choudhary. Evaluation of K-Means Data Clustering Algorithm on Intel Xeon Phi. *International Conference on Big Data* 2016

37. Diana Palsetia, William Hendrix, **Sunwoo Lee**, Ankit Agrawal, Wei-keng Liao, and Alok Choudhary. Parallel Community Detection Algorithm Using a Data Partitioning Strategy with Pairwise Subdomain Duplication. *International Conference on High Performance Computing (ISC)* 2016

38. **Sunwoo Lee**, Byung Kwan Jung, Minsoo Ryu, Seungwon Lee, Extending Component-based Approaches for Multi-threaded Design of Multiprocessor Embedded Software. *International Symposium on Object/Component/Service-Oriented Real-Time Distributed Computing* 2009

Pre-prints

1. Vincent-Daniel Yun, Junhyuk Jo, **Sunwoo Lee**, Weight Variance Amplifier Improves Accuracy in High-Sparsity One-Shot Pruning, *arXiv* 2025.

2. Hyuntak Shin, Aecheon Jung, Sungeun Hong, **Sunwoo Lee**, Dynamic Rank Adjustment for Accurate and Efficient Neural Network Training, *arXiv* 2025.

3. Jihyun Lim, Junhyuk Jo, Chanhyeok Ko, Young Min Go, Jimin Hwa, Sunwoo Lee, Biased Local SGD for Efficient Deep Learning on Heterogeneous Systems, *arXiv* 2025.

4. Zhenheng Tang, Xiaowen Chu, Ryan Yide Ran, **Sunwoo Lee**, Shaohuai Shi, Yonggang Zhang, Yuxin Wang, Alex Qiaozhong Liang, Salman Avestimehr, Chaoyang He, FedML Parrot: A Scalable Federated Learning System via Heterogeneity-Aware Scheduling on Sequential and Hierarchical Training. *arXiv* 2023.

5. Yue Niu, Saurav Prakash, Souvik Kundu, **Sunwoo Lee**, Salman Avestimehr, Federated Learning of Large Models at the Edge via Principal Sub-Model Training. *FL-NeurIPS* 2022.

6. Chaoyang He, Zhengyu Yang, Erum Mushtaq, **Sunwoo Lee**, Mahdi Soltanolkotabi, Salman Avestimehr, SSFL: Tackling Label Deficiency in Federated Learning via Personalized Self-Supervision. *arXiv* 2021

Invited Talks

- Department of Electrical Engineering at Hanyang University, South Korea: System-Efficient Federated Learning Methods, 2/26/2025
- Department of Immersive Media Engineering, Sunkyunkwan University, South Korea: Model Aggregation Strategies in Model Distributed Learning, 6/26/2024
- Department of Physics at Ajou University, South Korea: System-Aware Large-Scale Neural Network Training and its Applications, 3/20/2024
- Department of Computer Engineering at Gachon University, South Korea: Scalable Federated Learning Strategies on Real-world Edge Computing Environments, 11/18/2022
- Department of Electrical Engineering at Hanyang University, South Korea: Partial Model Training Strategies in Federated Learning, 10/04/2022
- U.S. Department of Energy, SciDAC, RAPIDS Institute, Tech Talk: Asynchronous I/O Strategy for Large-Scale Deep Learning Applications, 12/01/2021
- HDF5 User Group Meeting: A Case Study on Parallel HDF5 Dataset Concatenation for Scientific Data Analysis, 10/21/2021
- U.S. Department of Energy, SciDAC, RAPIDS Institute, Tech Talk: Communication-Efficient Local SGD for Scalable Deep Learning, 7/7/2021

Service

- Vice Chair of Computer Engineering Department at Inha University ~March 2025
- Program Committee of Association for the Artificial Intelligence (AAAI) 2021--2027
- Program Committee of IJCAI 2025
- Program Committee of NeurIPS 2023--2026
- Program Committee of KDD 2024--2026
- Program Committee of ICLR 2021, 2022, 2026, 2027
- Program Committee of ICML 2021, 2022, 2023, 2026
- Program Committee of CVPR 2026
- Program Committee of Engineering Applications of Artificial Intelligence (EAAI) 2024, 2025
- Reviewer in Expert Systems with Applications 2025

- Reviewer in Knowledge-based Systems 2024
- Reviewer in IEEE Transactions on Parallel and Distributed Computing 2023
- Reviewer in IEEE Transactions on Mobile Computing 2023, 2025
- Program Committee of International Conference on Artificial Intelligence and Statistics (AISTATS) 2021, 2022

Skills and Qualifications

Programming Language
C/C++, Python

Deep Learning Software Frameworks
TensorFlow, PyTorch, Caffe

Parallelization Libraries
MPI, OpenMP

I/O Libraries
MPI-I/O (ROMIO), HDF5, NetCDF